5.2 Evolution of Probability Measure

The solution operator describes how a function of the phase variables is mapped forward in time under the flow of the differential equation. An alternative ("dual") perspective is in terms of the evolution of the measure (or density) of points in phase space. We begin by summarizing a few basic principles needed to provide a foundation for working with probability measures.

5.2.1 Measure, Probability and Function Spaces

A probability distribution is assumed to be characterized by a *measure* on the state space of the problem which is the integral of a probability density function defined on the phase space of the problem that assigns nonnegative real numbers to each state. We outline the technical setting here. To gain a fuller understanding of the mathematical issues involved in this chapter, it may be helpful to refer to an introductory text on analysis such as [26, 317].

Assume, as usual, a system of N atoms with phase space having dimension m (typically $m = 2\mathrm{N}_c = 2\mathrm{N}_\mathrm{d} = 6\mathrm{N}$). Let $\mathcal{D} \subset \mathbb{R}^m$ be a suitable set comprising the collection of admissible phase space points $\mathbf{z}$.[1] A *density* on $\mathcal{D}$ is a function $\rho : \mathcal{D} \to \mathbb{R}$ such that (i) $\rho(\mathbf{z}) \geq 0, \mathbf{z} \in \mathcal{D}$, and (ii) $\int_\mathcal{D} \rho \mathrm{d}\omega < \infty$, where $\mathrm{d}\omega = \mathrm{d}z_1 \mathrm{d}z_2 \ldots \mathrm{d}z_m$ (i.e. $\int_\mathcal{D} \rho \mathrm{d}\omega$ is the integral of the measurable function ρ in the Lebesgue sense). A *probability density function* (or *p.d.f.*) on $\mathcal{D}$ is a density which is normalized so that

$$\int_\mathcal{D} \rho \mathrm{d}\omega = 1.$$

Any density (with positive total integral) can be normalized by dividing by $\int_\mathcal{D} \rho \mathrm{d}\omega$.

For any measurable subset $\mathcal{A}$ of $\mathcal{D}$, and probability density ρ, we define

$$\Pr(\mathcal{A}) \stackrel{\text{def}}{=} \int_\mathcal{A} \rho \mathrm{d}\omega.$$

[1] The technical requirement is that $\mathcal{D}$ be Lebesgue-*measurable*. There are some pathological sets for which this condition fails, but we will always assume the sets we are interested in are measurable. Open and closed sets in $\mathbb{R}^m$ are Lebesgue-measurable, for example a rectangular box. Sets which are constructed by taking countable unions or intersections of measurable sets are measurable. A suitable family of sets on which Lebesgue measure can be defined is the so-called Borel σ-algebra obtained by countable unions, intersections, or relative complements of open balls in $\mathbb{R}^m$.